

FANNO-Dev: 20 Key Claims with Quantitative Evidence

1	Scale	153,351 cleaned samples from 211K raw (27.4% rejection)	✓
2	Diversity	Vendi Score = 182.75 (highest known for instruction data)	✓
3	Multi-Pipeline	8 specialized synthesis pipelines, near-orthogonal (0.961 avg dist)	✓
4	Scaling Law	$V(N) = 55.4(1 - e^{-(N/745)}) + 109.0$, $R^2=0.996$	✓
5	Self-Inversion	2,274 exclusive domains via trajectory reversal (143× more than Complex QA)	✓
6	Selection	K-Center-Greedy +33% over random at N=500	✓
7	Multi-Turn	18,708 conversations, 11 patterns, 15 scenarios, 6.2 avg turns	✓
8	Quality	100% JSON valid, 0.01% refusal, 0 contamination	✓
9	Cost-Effective	2, 245total/(15.58/1K), Self-Inversion most efficient	✓
10	Tag Space	2,297 domains × 25 types, 4.1% utilization ceiling	✓
11	Embedding	>100 PCA components for 90% variance, high intrinsic dimensionality	✓
12	Complementarity	Leave-one-out: Doc QA +6.2%, Creative -3.2%	✓
13	Self-Inv Bridge	KNN isolation 0.224 (lowest), in all 20 clusters, highest entropy 3.636	✓
14	Spectral	Self-Inv slowest eigenvalue decay = most diverse	✓
15	Linguistic	Doc QA #1 composite (TTR + hapax + vocab), 32.7% exclusive vocabulary	✓
16	Mixing	Optimal: Self-Inv 3.2%→17.4%, Complex QA 35.4%→13.7%	✓
17	Hull Coverage	Self-Inv 87.6% of total PCA hull area (highest spatial coverage)	✓
18	Cross-Metric	TTR→Vendi $r=0.830$, Efficiency→Diversity $r=0.776$	✓
19	Robustness	Bootstrap CV: 3.2% at N=100, 1.1% at N=1000 (stable)	✓
20	Deliverables	98 figures, 25 tables, 50+ JSON reports, training recipe	✓

All claims verified with quantitative analysis. See paper_outline.md for figure/table cross-references.